

AWS Web Application Firewall (WAF)

Use Cases

- Pre-create security rules before application deployment
- Centralized security configuration
- Learning and understanding AWS WAF rule creation
- Future attachment to ALB, API Gateway, or CloudFront

Architecture Concept:

Incoming Request

↓

AWS WAF (Web ACL + Rules)

↓

[No resource attached yet]

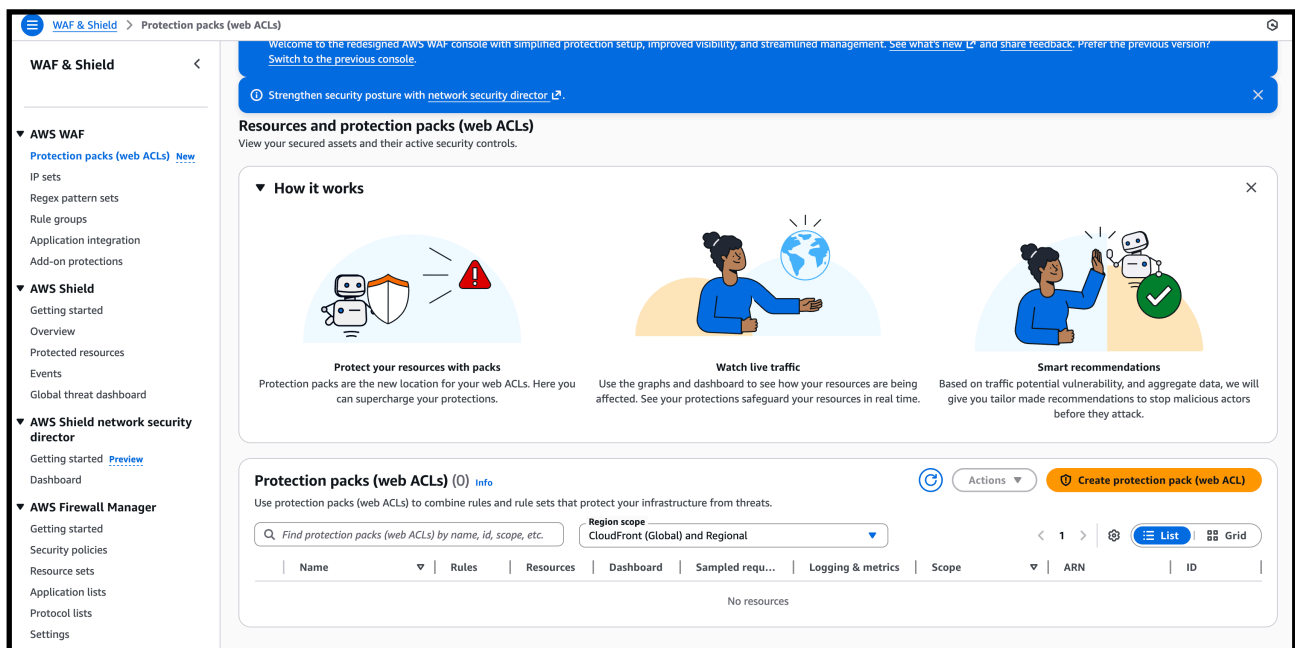
The firewall is created and ready but not enforcing traffic until associated.

Step-by-Step Execution

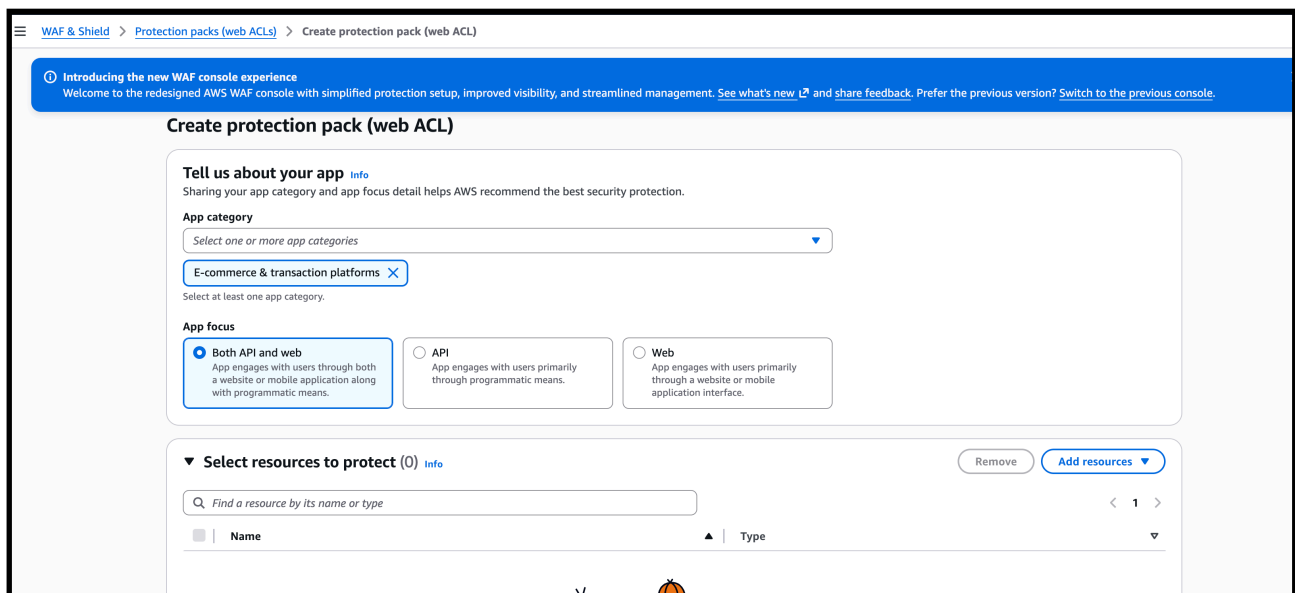
Step 1: Create Web ACL

AWS Console Path:

AWS Console → WAF & Shield → Web ACLs → Create web ACL



- App category selected: **E-commerce & transaction platforms**
- App focus selected: **Both API and web**



—> Provided application details by selecting app category as *E-commerce & transaction platforms* and app focus as *Both API and web*.

—> Select Resources to Protect

—> Choose Protection Pack Resource Type

Selected :

- **AWS Amplify application, CloudFront distribution, and CloudFront distribution tenant**

Choose protection pack (web ACL) resource types
The selected resource types can't be modified after creation.

☒ AWS Amplify application, CloudFront distribution, and CloudFront distribution tenant

☐ Amazon API Gateway REST API, Application Load Balancer, AWS AppSync GraphQL API, Amazon App Runner service, Amazon Cognito user pool, and AWS Verified Access

▼ Choose initial protections [Info](#)
AWS WAF offers various protection packages of WAF rules. AWS WAF rule configurations are based on security best practices. You should validate the rule configurations that are best for you. You can also choose individual rules instead of packages.

Recommended

☒ **Recommended rules for you**
Enables recommended rules for the selected application categories and app focus.

Everything in Essentials
and the following

- ✓ Rate limit for GET requests [Count](#)
- ✓ Rate limit for POST, PUT, DELETE requests [Count](#)
- ✓ Amazon IP reputation list [Default](#)
- ✓ Anonymous IP list protection [Default](#)
- ✓ Managed IP DDoS protection [Default](#)
- ✓ Bot Control protection [Count](#)

Essentials

☐ **Essential rules**
Enables essential rules for the selected application categories and app focus.

- ✓ Layer 7 anti-DDoS [Count](#)
- ✓ IP allowlist [Allow](#)
- ✓ IP blocklist [Block](#)
- ✓ Geographic restriction [Block](#)
- ✓ Global rate limit [Count](#)
- ✓ Body size restriction rule [Count](#)
- ✓ AWS core rule set [Default](#)
- ✓ Known bad inputs protection [Default](#)

You build it

☐ **Build your own pack from all of the protections AWS WAF offers**
Select and customize rules to fit your needs.

- ✓ You choose which rules to apply

—> Selected CloudFront and Amplify related resource types for the Web ACL.

—> Choose Initial Protections

What you did

- Selected **Recommended rules for you**
- Kept default recommended protections enabled

—> Selected AWS recommended protection rules to secure the application.

Name and Describe Protection Pack

The screenshot shows the 'Name and describe' step in the AWS WAF console. It includes a 'Name' field with the value 'my-waf-firewall' and a 'Description' field which is empty. A note states: 'Name cannot be changed later. Valid characters are letters, numbers, and hyphens (-)'. Another note says: 'The description can have up to 256 characters.' At the bottom, there is a 'Customize protection pack (web ACL) - optional' link and two buttons: 'Cancel' and 'Create protection pack (web ACL)'.

Name entered: **my-waf-firewall**

Description: optional (left empty)

—> Provided the name *my-waf-firewall* for the AWS WAF protection pack.

Create Protection Pack

- Reviewed all settings
- Clicked **Create protection pack (web ACL)**

The screenshot shows the AWS WAF console interface. At the top, a green banner indicates 'Successfully created protection pack (web ACL) my-waf-firewall.' Below this, the 'Resources and protection packs (web ACLs)' section shows the 'my-waf-firewall' pack. The 'How it works' section includes three cards: 'Protect your resources with packs', 'Watch live traffic', and 'Smart recommendations'. The 'Protection packs (web ACLs) (1)' table lists the 'my-waf-firewall' pack with 19 rules, a 'Manage' button, and a 'View' button. The right sidebar shows the 'my-waf-firewall' details, including its ARN, ID, and a list of actions: 'Manage details', 'Manage rules', 'Manage resources', 'View dashboard, logs and sampled requests', and 'Configure logging and sampled requests'.

—> **Protection Pack Created Successfully**

- Success message displayed
- Protection pack status active
- Rules enabled

Conclusion :

—> AWS WAF protection pack *my-waf-firewall* was created successfully.